Title: **An Automatic Pipeline for Sulcal Feature Extraction in Fetal Surface MRI**

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Introduction: Malformations of cortical development (MCD) in the prenatal period can disrupt normal folding (e.g. isolated wide sulci) and lead to cognitive deficits (e.g. autism) [1]. Folding develops rapidly in late gestation, and deviations from normative patterns may signal abnormal development. Fetal MRI is valuable for diagnosing MCDs, but individual variability in folding complicates: (A) localisation of folds; and (B) the biological definition of "normative" development.

Surface-based methods model the cortex as a 2D manifold, enabling accurate sulcal localization and extraction of morphometric features (i.e. sulcal depth and length). In this abstract, we therefore present a holistic framework for automated analysis of fetal sulcal formation. This starts with deep learning-based fetal tissue segmentation and surface extraction, then performs automated parcellation of sulcal regions, and finally tracks sulcal fundi to extract multiple morphometric features, including sulcal surface area (A_i), sulcal length (L_i), sulcal depth (D_i), sulcal curvature (C_i), and sulcal span ($S_i = A_i/L_i$), with $i = \{1,\dots 29\}$ different primary and secondary sulci. Based on their typical fetal emergence time window (around 17-23 week, 24-27 week, and >=28 week, respectively), the structures were grouped into early-forming (e.g. the lateral fissure, calcarine sulcus, parieto-occipital fissure, and cingulate/central sulci), intermediate-forming (e.g. intermediate folds included pre/postcentral, superior frontal, superior temporal, and intraparietal sulci), and late-forming folds (e.g. later/variable folds included inferior frontal/temporal, medial parietal, lateral occipital, occipito-temporal lateral, orbitofrontal, paracentral, and posterior/paracingulate).

Methods: We analyzed 184 fetal dHCP scans (GA: 24.0–38.3 wks; mean ± SD: 30.2 ± 3.6) [2], acquired at a Philips Achieva 3T system and reconstructed at 0.5 mm isotropic SR. Preprocessing included BET (brain extraction), bias field correction, BOUNTI (tissue segmentation) and CoTan (surface reconstruction) (Figure 1A). We hand-drew the borders of 29 ROIs for each of the weekly fetal cortical surface templates, generated according to the methods of Bozek et al 2018, with sulci named according to the nomenclature described in Snyder at al 2024 [3]. These were then mapped to individuals via their (nearest in age) template using MSMsulc (Fig 1B1) and visually checked to ensure their accuracy.

Sulcal features were then derived from the inner cortical (or white) surface, by first thresholding surface curvature maps to only retain the regions of interest (ROI) with most negative curvature (Fig 1B2); next most convex points within this region were selected - with the furthest pair designated as sulcal end points. Finally, we applied a curvature gradient weighted Dijkstra's algorithm (a commonly used graph optimization method) to find the shortest consistent path connecting the ends at the locations where curvature was minimized. In this way, we identified fundi as curves lying within areas of extreme curvature (Fig 1B3). From this it was possible to estimate the sulcal depth (deepest point), sulcal area (total vertex area of ROI), the mean curvature, sulcal length and sulcal span.

To validate our work, we plot the normative trajectories of development for each of these features using Gaussian Process regression. These results are shown for two early formed (e.g. Parieto-Occipital Fissure, Central Sulcus), one intermediate (e.g. Superior Temporal Sulcus) and two mid-to-late formed sulci (e.g. Medial Parietal Fissure, Posterior Cingulate Sulcus).

Results: GP model fit results showed that each sulci displays a unique trajectory of development at each metric. Figure 2 showed five representative sulci spanning earlier- to later-maturing profiles (parieto-occipital fissure, central sulcus, superior temporal sulcus, posterior cingulate sulcus, and medial parietal sulcus) across six geometric features. Across sulci, the GP captured smooth, biologically plausible growth patterns with metric-specific timing: curvature and depth typically showed earlier and more time-confined acceleration, whereas length tended to change more gradually and often lacked an internal peak within the sampled age range (boundary-limited), consistent with continued maturation toward term-equivalent age. Importantly, sulci commonly considered earlier-forming (e.g., central sulcus and parieto-occipital fissure) tended to exhibit earlier peak slope in curvature-related metrics than later-maturing sulci (e.g., medial parietal sulcus and superior temporal sulcus), supporting sensitivity of the proposed features to developmental timing.

Conclusion: Our results demonstrate that this automatic sulcal feature analysis framework captures known developmental patterns [4,5] and broadens clinical use by enabling automated, scalable early cortical maturation analysis.

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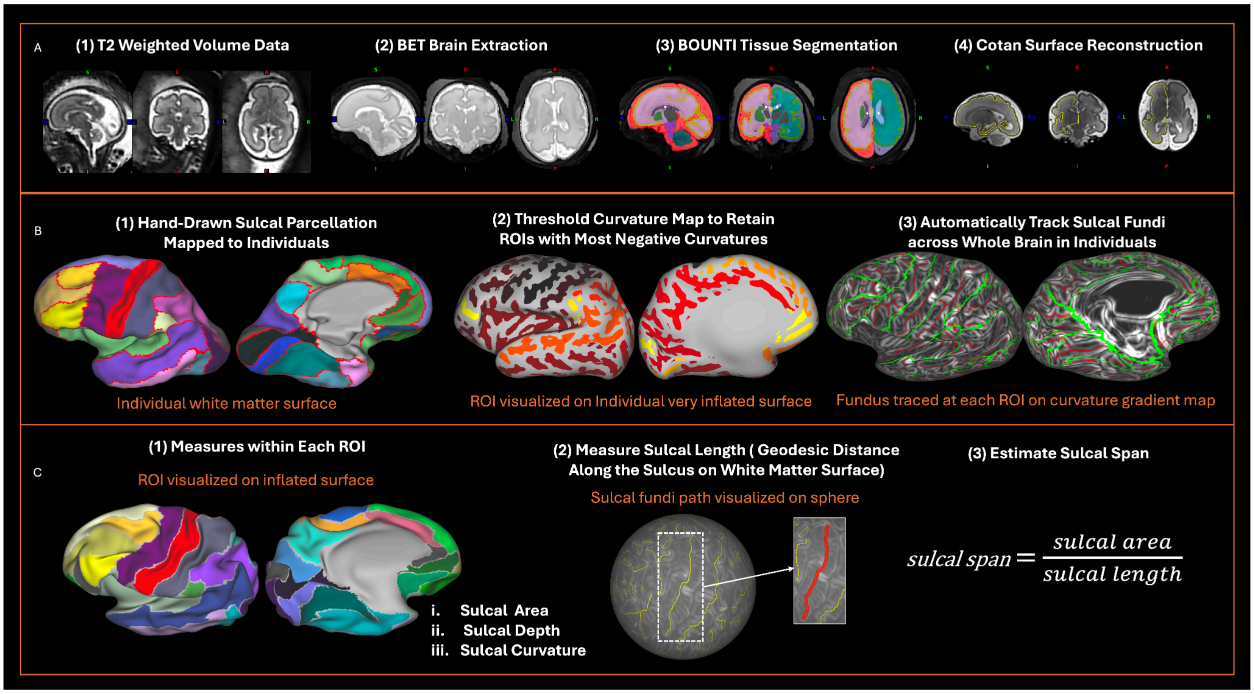

Figure 1. Overview of the proposed pipeline for normative modelling of fetal sulcal development. (A) preprocessing: brain extraction (BET), tissue segmentation (BOUNTI), and deep learning based cortical surface reconstruction (CoTan). (B) shows the semi-automated definition of sulcal ROIs on individual cortical surfaces via template-to-individual mapping of manually defined template ROIs using surface-based registration (MSM). (C) shows the automated extraction of sulcal morphological features derived from surface-based anatomical landmarks.

Example of normative modeling of sulcus at each measurement shows inter-individual variance

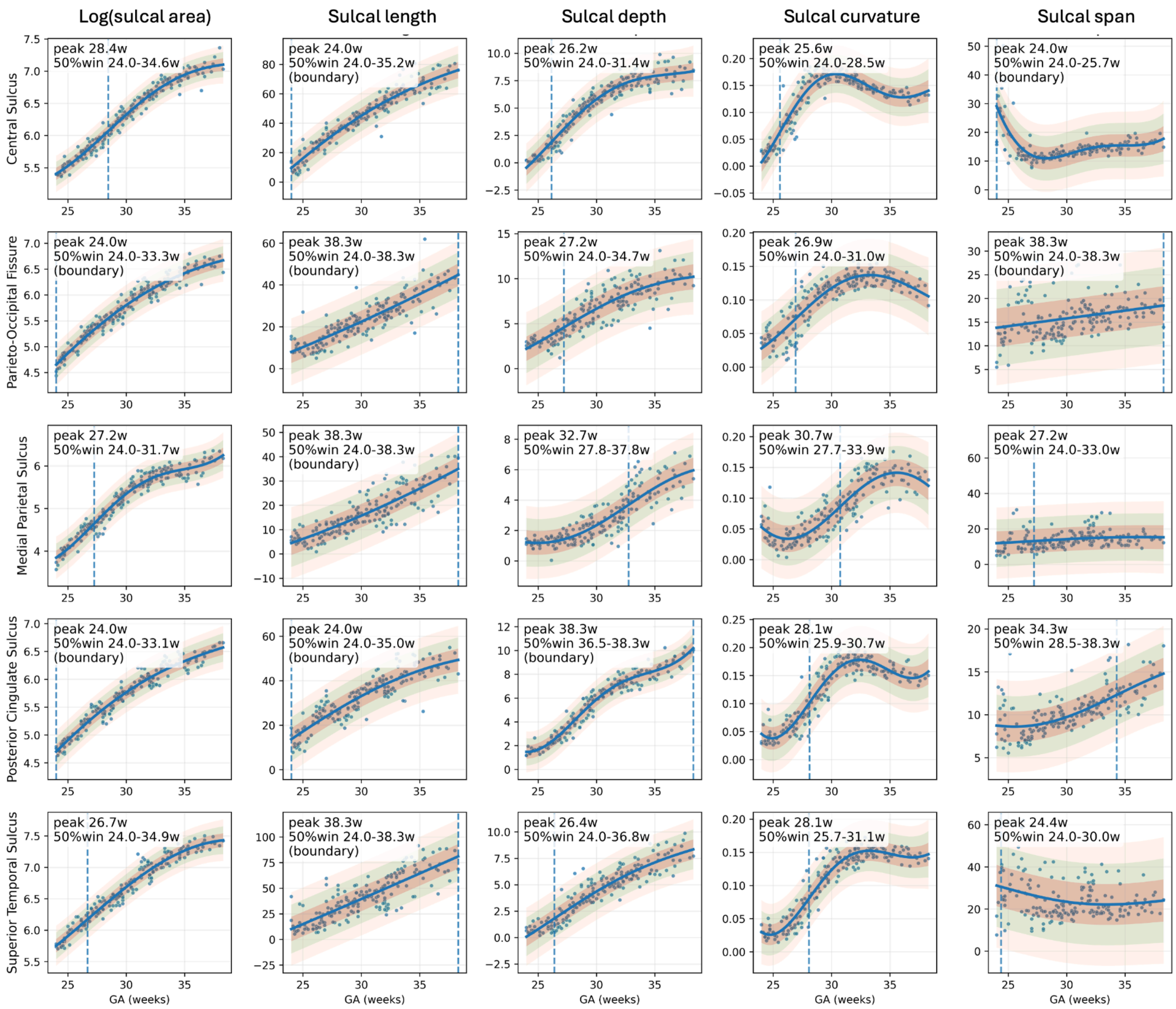

Figure 2. illustrates normative developmental trajectories of key features of five representative sulci including Central Sulcus, Parieto-Occipital Fissure, Medial Parietal Fissure, Posterior Cingulate Sulcus and Superior Temporal Sulcus. Features include sulcal area, sulcal length, sulcal depth, sulcal curvature, and sulcal span (sulcal area/sulcal length).

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